

GHCNh VERSION UPDATE: From v1.0.1 (previous) to v1.1.0 (new)

March 10, 2026

Changes from previous version:

- 1) New variables added (19 cloud/sky and 2 high-resolution precipitation elements)
- 2) Some sky variables were removed and replaced
- 3) Some data abbreviations, measurement codes, and report type codes have been changed
- 4) Header/column changes

1) 21 new/replaced elements (21 cloud/sky and 2 high-resolution precipitation variables) in the new version

- sky_condition
- sky_condition_baseht
- ceiling_height
- sky_cover_layer_1 (formerly sky_cover_1)
- sky_cover_layer_2 (formerly sky_cover_2)
- sky_cover_layer_3 (formerly sky_cover_3)
- sky_cover_layer_4
- sky_cover_layer_baseht_1 (formerly sky_cover_baseht_1)
- sky_cover_layer_baseht_2 (formerly sky_cover_baseht_2)
- sky_cover_layer_baseht_3 (formerly sky_cover_baseht_3)
- sky_cover_layer_baseht_4
- sky_cover_summation_1
- sky_cover_summation_2
- sky_cover_summation_3
- sky_cover_summation_4
- sky_cover_summation_baseht_1
- sky_cover_summation_baseht_2
- sky_cover_summation_baseht_3
- sky_cover_summation_baseht_4
- precipitation_5_minute
- precipitation_15_minute

All new/existing elements continue to have associated columns for codes, etc:

precipitation_15_minute_Measurement_Code
precipitation_15_minute_Quality_Code
precipitation_15_minute_Report_Type
precipitation_15_minute_Source_Code
precipitation_15_minute_Source_Station_ID

2) These sky cover elements were removed and replaced in the new version, along with the expanded set of sky elements shown above.

sky_cover_1
sky_cover_2
sky_cover_3
sky_cover_baseht_1

sky_cover_baseht_2
sky_cover_baseht_3

3) Cleaner/truncated data/code fields in the new version (these are just a few examples). See the [GHCHh readme documentation](#) for more details.

- FM15-METAR is now FM15 (report_type)
- AUTO_4-USA is now AUTO (report_type)
- C-Calm is now C (measurement_code_1 for wind elements)
- FG:45-Fog- is now FG:45 (present weather codes will be abbreviated)
- 9-Missing is now be blank fields

4) See more details on header and element differences here: [ghcnh-columns.pdf](#). Be sure to scroll down to view all changes.